

Loke Kum Yew

Full-Stack Engineer & AI Systems Architect

Specialising in Scalable Web Platforms & AI-Augmented Development

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Career Interest

Full-Stack Engineer with First-Class Honours, specialising in **AI-augmented development workflows**. I partner with startups and enterprises to architect and deliver robust web platforms and cloud infrastructure, and I am an expert in **stabilising fragile legacy web platforms**, building high-performance **offline-first mobile architectures**, and **cross-platform desktop tooling**. Proven track record delivering high-stakes projects, from bespoke enterprise event systems to venture-backed SaaS acceleration.

Technical Skills

- **Web Development (Primary):** TypeScript, React, Next.js, Node.js, Python, Tailwind CSS, PHP.
- **Cloud & Infrastructure:** AWS (Lambda, Cognito, S3, CloudFront), Supabase (PostgreSQL/RLS), MongoDB.
- **Mobile & Desktop Development:** React Native (Expo), Offline-First Architecture (WatermelonDB/SQLite), Electron (Cross-Platform Desktop).
- **AI Integration:** Prompt Engineering, AI-driven development (Claude Code, Cursor, Cline, GitHub Copilot).
- **Tools & DevOps:** CI/CD (GitHub Actions), Docker, Vercel, Linear, Figma.

Professional Experience

Principal Software Consultant | Loke Software

June 2024 – Present | Remote (Based in Bangkok/Singapore)

Strategic technical partner for international startups and enterprises. Manage a high-stakes portfolio of engagements focused on system stabilization, cloud architecture, and offline-first mobile development.

Venture-Backed Travel SaaS (USA) | Lead Web Engineer

- **Infrastructure & Security:** Spearheaded forensic audit and remediated vulnerabilities via refactored AWS Lambda and Cognito auth flows.
- **Modernization:** Led Node 20 migration and standardized CI/CD environment management to ensure 99.9% production uptime.
- **Systems Architecture:** Re-engineered payment and registration modules for high-concurrency traffic and complex multi-role permissions.
- **Process Leadership:** Implemented automated GitHub Actions and formal QA protocols to mitigate technical debt and increase velocity.

Enterprise Event Registration Platform | Systems Architect

- **Hybrid Architecture:** Architected an installable desktop app switching between fully offline (SQLite) and online (Supabase) modes for maximum event-day reliability.
- **Live Deployment:** Checked in 400+ attendees across a 2-day AI Engineer conference using QR scanning and instant badge printing from a custom template designed for the event.
- **Admin Console:** Built a real-time admin console for attendee management, per-event configuration (search, QR toggles, PDPA modes), and a drag-and-drop badge editor.

Interactive UV Print Experience Platform | Full-Stack Developer

- **Design-to-Print Kiosk:** Architected a cloud-hosted kiosk platform letting attendees design custom merchandise via a touch-friendly canvas editor on tablet kiosks, with a scan-to-upload mobile bridge for login-free photo uploads.
- **Hardware Integration:** Engineered a background sync agent auto-feeding finished, CMYK-ready designs into the UV printer's RIP software — a hardware-agnostic platform across concurrent events, zero manual file handling.

Scientific Data Platform | Mobile Systems Architect

- **Local-First Persistence:** Developed a high-performance React Native (Expo) application using WatermelonDB (SQLite) for 100% offline functionality.
- **Data Engineering:** Designed a custom pull-only sync engine orchestrating large-scale data and media transfers from Supabase.
- **Iterative Feature Delivery:** Shipped user authentication, an admin web interface for species metadata, and data export across successive milestone releases.

AI SaaS Platform | Prompt Architecture

- **Technical Audit & Reasoning Engineering:** Conducted forensic audits of the generation platform, identifying systemic architectural flaws and data bottlenecks; architected V2 prompt reasoning chains and structured output schemas to ensure deterministic model behavior and minimize LLM hallucination.

Selected Web Projects

Interactive SG60 Magic Ball | Next.js & Supabase

- **Engineered:** a high-performance dynamic UI with complex CSS-driven holographic projections, backed by a serverless architecture with Row Level Security (RLS) for enterprise-grade data protection.

[Live Site](#) | [GitHub](#)

Email Verification SaaS Engine | Python & Redis

- **Designed:** a cost-optimized, multi-stage email validation engine using Redis caching, cutting paid API usage and operational costs by 36%.

Education

Nanyang Technological University (NTU) | Singapore

Bachelor of Engineering (Mechanical & Mechatronics)

- **Honours:** First Class Honours (Distinction), GPA: 4.91/5.00
- **Engineering Foundation:** Strong foundation in logic, control systems, and complex system design.

Meta React Native Certification | Coursera

- **Specialization:** Cross-platform mobile development and UI/UX implementation.